Summation Based on Exponential Series including sine and cosine Series.

$$e^x = 1 + x + \cancel{\frac{x}{1!}} + \frac{x^2}{2!} + \cdots$$

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

$$\sin x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Q. Sum the series $1 + \frac{c^2 \cos 2\theta}{2!} + \frac{c^4 \cos 4\theta}{4!} + \cdots$

$$C = 1 + \frac{c^2 \cos 2\theta}{2!} + \frac{c^4 \cos 4\theta}{4!} + \cdots$$

$$S = \frac{c^2 \sin 2\theta}{2!} + \frac{c^4 \sin 4\theta}{4!} + \cdots$$

$$C + iS = 1 + \frac{c^2}{2!}(\cos 2\theta + i \sin 2\theta) + \frac{c^4}{4!}(\cos 4\theta + i \sin 4\theta) + \cdots$$

$$= 1 + \frac{c^2}{2!}(e^{i\theta})^2 + \frac{c^4}{4!}(e^{i\theta})^4 + \cdots$$

$$= \cosh(ce^{i\theta})$$

$$= \cosh(c(\cos\theta + i\sin\theta))$$

$= \cosh\left(c\cos\theta + ic\sin\theta\right)$ $\left(\because \cosh x = \cos ix\right)$

$= \cos i\left(\cancel{\cos\theta + i\sin\theta}\ c\cos\theta + ic\sin\theta\right)$

$= \cos\left(ic\cos\theta + i^2 c\sin\theta\right)$

$= \cos\left(ic\cos\theta - c\sin\theta\right)$

$= \cos(ic\cos\theta).\cos(c\sin\theta) + \sin(ic\cos\theta).\sin(c\sin\theta)$

$C + iS = \cosh(c\cos\theta).\cos(c\sin\theta) + i\sinh(c\cos\theta).\sin(c\sin\theta)$

$\Rightarrow C = \cosh(c\cos\theta).\cos(c\sin\theta)$

Q. $\sin\alpha + c\sin(\alpha+\beta) + \frac{c^2}{2!}\sin(\alpha+2\beta) + \cdots$

$C = \cos\alpha + c\cos(\alpha+\beta) + \frac{c^2}{2!}\cos(\alpha+2\beta) + \cdots$

$S = \sin\alpha + c\sin(\alpha+\beta) + \frac{c^2}{2!}\sin(\alpha+2\beta) + \cdots$

$C + iS = (\cos\alpha + i\sin\alpha) + c\left(\cos(\alpha+\beta) + i\sin(\alpha+\beta)\right) + \frac{c^2}{2!}\left(\cos(\alpha+2\beta) + \sin(\alpha+2\beta)\right) + \cdots$

$= e^{i\alpha} + c\,e^{i(\alpha+\beta)} + \frac{c^2}{2!}\,e^{i(\alpha+2\beta)}$

$= e^{i\alpha}\left(1 + c\,e^{i\beta} + \frac{c^2}{2!}\,e^{i2\beta} + \cdots\right)$

$= e^{i\alpha}.e^{cie^{\beta}}$

$$= e^{i\alpha} \, e^{c[\cos\beta + i\sin\beta]}$$

$$= e^{i\alpha} \, e^{c\cos\beta} \cdot e^{ic\sin\beta}$$

$$= e^{c\cos\beta} \cdot e^{i(\alpha + c\sin\beta)}$$

$$= e^{c\cos\beta}\left[\cos(\alpha + c\sin\beta) + i\sin(\alpha + c\sin\beta)\right]$$

$$\Rightarrow S = e^{c\cos\beta} \, \sin(\alpha + c\sin\beta)$$

Q. $1 - \cos\alpha\cos\beta + \dfrac{\cos^2\alpha}{2!}\cos 2\beta - \dfrac{\cos^3\alpha}{3!}\cos 3\beta + \cdots$

$$S = -\cos\alpha\sin\beta + \frac{\cos^2\alpha}{2!}\sin 2\beta - \cdots$$

$$\cancel{C = 1 + \sin\beta + \frac{\cos^2\alpha}{2!}\sin 2\beta + \frac{\cos^3\alpha}{3!}\sin 3\beta + \cdots}$$

$$C + iS = 1 - \cos\alpha\,(e^{i\beta}) + \frac{\cos^2\alpha}{2!}e^{i2\beta} - \frac{\cos^3\alpha}{3!}e^{i3\beta} + \cdots$$

$$= e^{-\cos\alpha(\cos\beta + i\sin\beta)}$$

$$= e^{-\cos\alpha\cos\beta} \cdot e^{-i\cos\alpha\sin\beta}$$

$$= e^{-\cos\alpha\cos\beta}\left[\cos(\cos\alpha\sin\beta) - i\sin(\cos\alpha\sin\beta)\right]$$

$$C = e^{-\cos\alpha\cos\beta} \cos(\cos\alpha\sin\beta)$$

HW Q. $\cos\alpha + c\cos(\alpha+\beta) + \frac{c^2}{2!}\cos(\alpha+2\beta) + \cdots$

Q. Sum the series $\sin\alpha - \frac{\sin(\alpha+2\beta)}{2!} + \frac{\sin(\alpha+4\beta)}{4!} + \cdots$

$S =$

$C = \cos\alpha - \frac{\cos(\alpha+2\beta)}{2!} + \frac{\cos(\alpha+4\beta)}{4!} + \cdots$

$C+iS = (\cos\alpha + i\sin\alpha) - \frac{1}{2!}\left(\cos(\alpha+2\beta) + i\sin(\alpha+2\beta)\right] +$

$\frac{1}{4!}\left[\cos(\alpha+4\beta) + i\sin(\alpha+4\beta)\right] + \cdots$

$= e^{i\alpha} - \frac{e^{i(\alpha+2\beta)}}{2!} + \frac{1}{4!}e^{i(\alpha+4\beta)} - \cdots$

$= e^{i\alpha}\left[1 - \frac{e^{i2\beta}}{2!} + \frac{1}{4!}e^{i4\beta} + \cdots\right]$

$= e^{i\alpha}\cos(e^{i\beta})$

$= e^{i\alpha}\left[\cos(\cos\beta + i\sin\beta)\right]$

$= e^{i\alpha}\left[\cos(\cos\beta)\cos(i\sin\beta) - \sin(\cos\beta)\sin(i\sin\beta)\right]$

$= e^{i\alpha}\left[\cos(\cos\beta)\cosh\sin\beta - \sin(\cos\beta)\, i\sinh\sin\beta\right]$

$= (\cos\alpha + i\sin\alpha)\left[\cos(\cos\beta)\cosh(\sin\beta) - i\sin(\cos\beta)\sinh(\sin\beta)\right]$

$\Rightarrow S = -\cos\alpha\sin(\cos\beta)\sinh(\sin\beta) + \sin\alpha\left(\cos(\cos\beta)\cosh(\sin\beta)\right)$

HW Q. $\cos\alpha - \dfrac{\cos(\alpha+2\beta)}{3!} + \dfrac{\cos(\alpha+4\beta)}{5!} - \cdots$